

20CS4T02 – Design and Analysis of Algorithms

UNIT 1: Algorithm Analysis

Introduction

Algorithm analysis evaluates the efficiency of algorithms in terms of time and space complexity. It helps in selecting the best algorithm for a given problem.

This unit introduces asymptotic analysis and complexity measures.

Detailed Explanation

Time Complexity

Time complexity measures the execution time of an algorithm.

Big O notation is used to express complexity.

Space Complexity

Space complexity measures memory usage.

Asymptotic Notations

Notations such as Big O, Omega, and Theta describe algorithm performance.

Applications

Algorithm analysis is used in optimizing software and solving computational problems efficiently.

Summary

This unit introduces complexity analysis and performance evaluation.

UNIT 2: Divide and Conquer

Introduction

Divide and conquer is a strategy that breaks a problem into smaller subproblems, solves them, and combines results.

This unit focuses on its applications.

Detailed Explanation

Concept and Approach

Problems are divided recursively into smaller parts.

Examples

Algorithms such as merge sort and quick sort use this technique.

Applications

Used in sorting, searching, and computational geometry.

Summary

This unit explains divide and conquer strategy.

UNIT 3: Greedy Algorithms

Introduction

Greedy algorithms make locally optimal choices to achieve a global optimum.

Detailed Explanation

Concept

At each step, the best immediate choice is made.

Examples

Algorithms like Kruskal's and Prim's are based on greedy approach.

Applications

Used in optimization problems.

Summary

This unit introduces greedy methods.

UNIT 4: Dynamic Programming

Introduction

Dynamic programming solves problems by breaking them into overlapping subproblems.

Detailed Explanation

Concept

Stores results of subproblems to avoid recomputation.

Examples

Fibonacci sequence and knapsack problem.

Applications

Used in optimization and decision-making problems.

Summary

This unit explains dynamic programming techniques.

UNIT 5: NP-Complete Problems

Introduction

NP-complete problems are complex problems for which no efficient solution is known.

Detailed Explanation

Classes of Problems

P, NP, and NP-complete classes.

Examples

Travelling Salesman Problem.

Applications

Used in theoretical computer science.

Summary

This unit introduces complexity classes.